Lecture 1: Data Structure and Baseline Forecasting

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Roadmap (Lecture 1)

Part A: Data structure and forecasting setup

- Three data structures: cross-sectional, time series, panel
- Forecast object, forecast horizon, information set, conditional expectation
- Unobserved component decomposition, log return, and demeaned log return
- Regression foundations and the bridge to autoregression

Part B: Baselines and evaluation

- Baseline models: naive, mean, random walk, AR(1)
- AR(p), stationarity, and ARIMA overview
- In-sample vs out-of-sample, R^2, and forecast error metrics

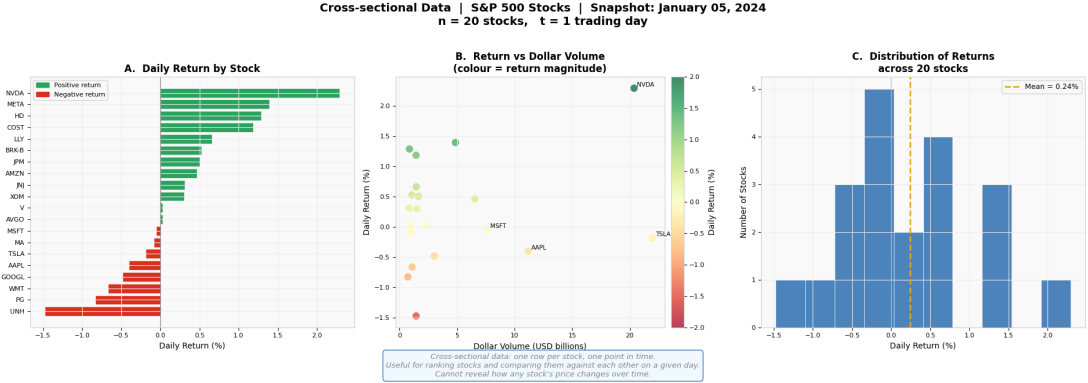
Cross-Sectional Data $(n > 1,\ t = 1)$

- Observations come from many units at one point in time.
- Typical examples: household income survey, firm-level balance-sheet snapshot.
- Useful for comparing differences across entities.

Source links

- Definition: `https://en.wikipedia.org/wiki/Cross-sectional_data`
- Figure file: `graph/block1_cross_sectional.png`

Cross-Sectional Data $(n > 1,\ t = 1)$



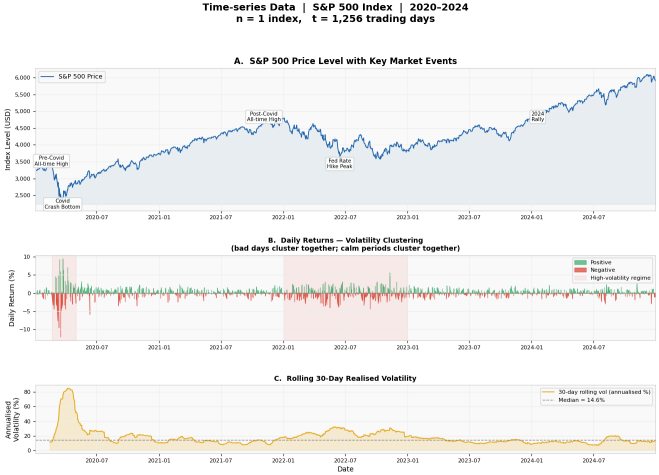
Time-Series Data $(n = 1,\ t > 1)$

- Observations track one unit across multiple time periods.
- Typical examples: monthly sales of one store, daily stock index levels.
- Core data structure for forecasting future values.

Source links

- Definition: `https://en.wikipedia.org/wiki/Time_series`
- Figure file: `graph/block2_time_series.png`

Time-Series Data $(n = 1,\ t > 1)$



Time-series data: one entity, many time points.
Reveals long-run trends, volatility regimes, and the impact of macro events.

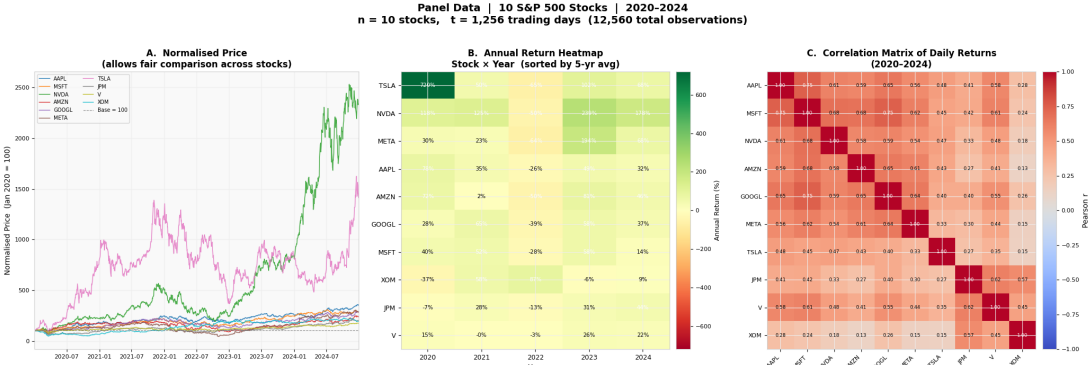
Panel Data $(n > 1,\ t > 1)$

- Observations track many units across multiple periods.
- Typical examples: quarterly sales for many firms, annual GDP for many countries.
- Combines cross-sectional heterogeneity with time dynamics.

Source links

- Definition: `https://en.wikipedia.org/wiki/Panel_data`
- Figure file: `graph/block3_panel.png`

Panel Data $(n > 1,\ t > 1)$



Forecast Object and Information Set

Forecast object

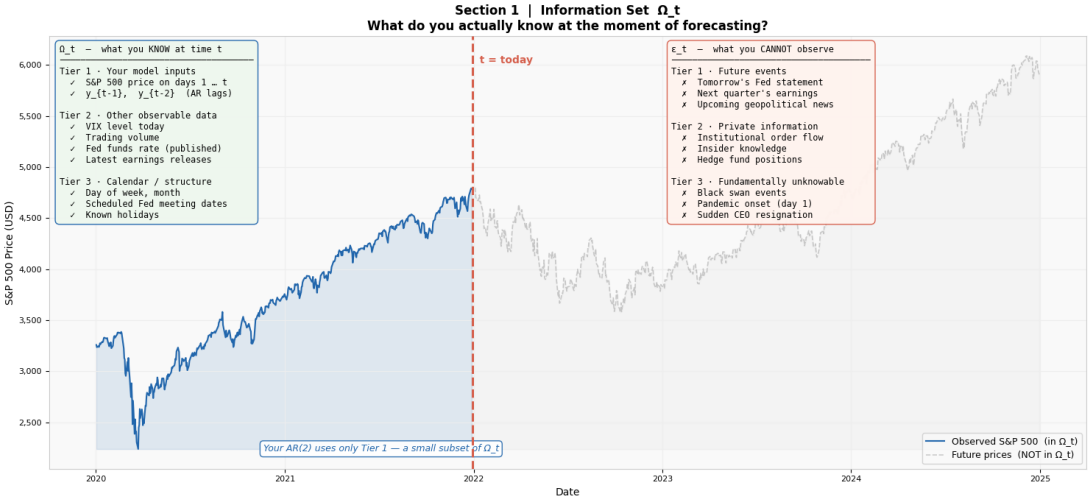
$$\hat{y}_{t+h|t} = f(\mathcal{I}_t)$$

- $\hat{y}_{t+h|t}$: forecast made at time t for period $t+h$
- h: forecast horizon
- $\mathcal{I}_t$: information available at time t

One-step-ahead example

$$\hat{y}_{t+1|t} = f(y_t, y_{t-1}, \ldots, y_{t-p+1})$$

Information Set Figure



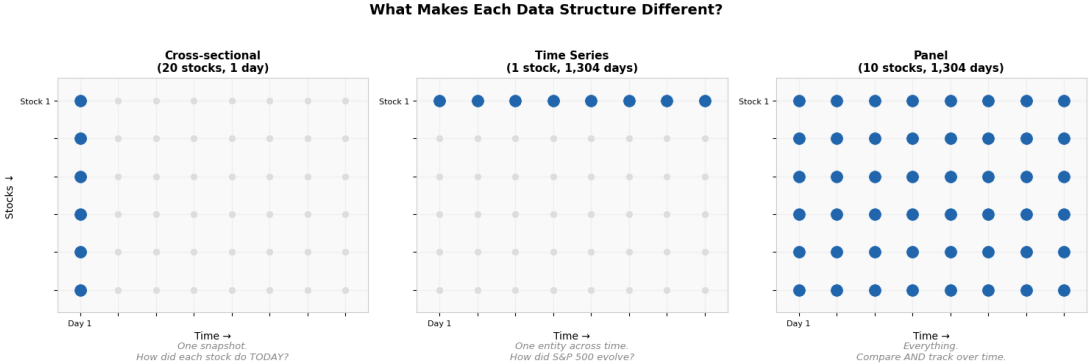
Conditional-Expectation Benchmark

Under squared-error loss, the statistical ideal forecast is:

$$\hat{y}_{t+1|t} = E(y_{t+1} \mid \mathcal{I}_t)$$

- This minimizes expected squared forecast error.
- Practical models approximate this benchmark.
- Out-of-sample performance is the key evaluation criterion.

Conditional-Expectation Benchmark Figure



Regression Forecasting: General Setup

$$y_t = \beta_0 + \beta_1 x_t + u_t$$

- y_t: target variable at time t
- x_t: explanatory variable at time t
- u_t: unexplained component (error term)

Forecasting starts from a regression structure and then changes what enters x_t.

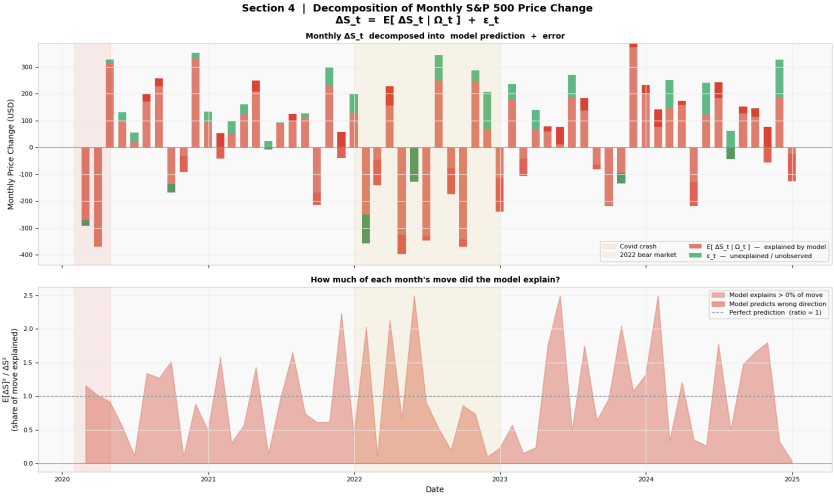
Unobserved Components and Cyclical Time Series

$$y_t = \mu_t + c_t + \varepsilon_t$$

- μ_t: slow-moving level/trend component
- c_t: cyclical component (medium-run fluctuations)
- ε_t: irregular shock (unobserved noise)

Observed data mix latent components. In market data, index levels (e.g., S&P 500) can be read as trend plus cyclical deviations.

SP500 Decomposition Figure



Log Return: Meaning and Why It Matters

Definition for price series P_t

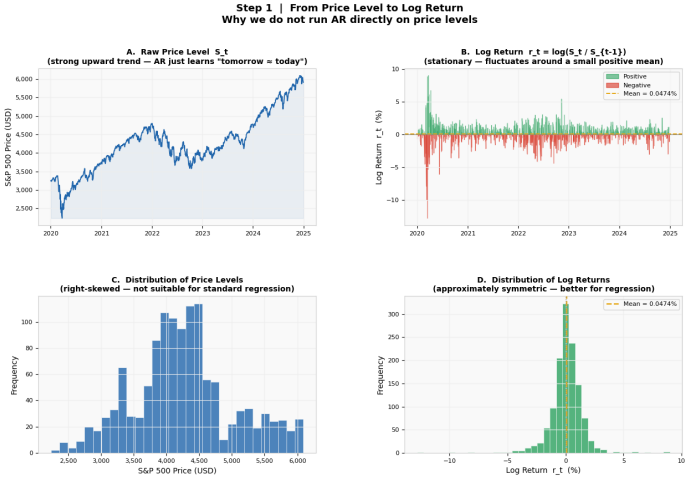
$$r_t = \log(P_t) - \log(P_{t-1}) = \log\left(\frac{P_t}{P_{t-1}}\right)$$

- r_t: continuously compounded one-period return.
- $r_t = 0.01$ *is approximately a* 1% *one-period return.*

For forecasting, log returns are useful because:

- Converts multiplicative growth into additive changes.
- Multi-period log returns add directly across time.
- Usually more stable than raw price levels (often closer to stationarity).

Log Return Figure



Log returns remove the price-level trend and make returns comparable across time.
In macroeconomics you would additionally deflate by CPI to get real returns — here we focus on the statistical transformation.

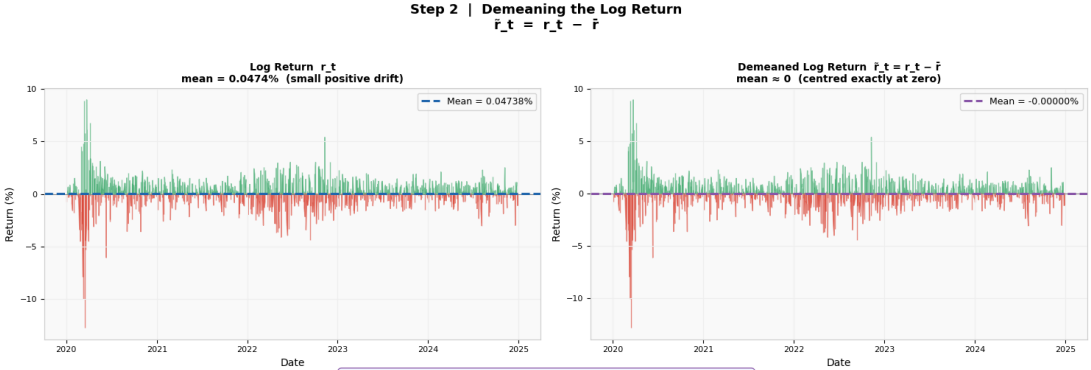
Demeaned Log Return and Cyclical Deviation

$$\tilde{r}_t = r_t - \bar{r}, \qquad \bar{r} = \frac{1}{T}\sum_{t=1}^{T} r_t$$

- Centers returns around zero to highlight above/below-average movements.
- Positive $\tilde{r}_t$: above-average return; negative: below-average return.
- Useful for visualizing cycle-like variation in return dynamics.

For real-time prediction, use rolling or expanding means to avoid look-ahead bias.

Demeaned Log Return Figure



Demeaning shifts the series so it is centred at zero.
The AR model can then focus on the dynamics — not on estimating the mean itself.
Standard deviation is unchanged: only the location shifts.

Linear Regression with Multiple Predictors

$$y_t = \beta_0 + \beta_1 x_{1,t} + \beta_2 x_{2,t} + \cdots + \beta_k x_{k,t} + u_t$$

- "Linear" means linear in parameters (β's).
- We estimate unknown coefficients from historical data.
- We plug in known/future predictor values to form forecasts.

Time-series forecasting replaces external x's with lagged y's.

Regression: $y \sim x$

Specification

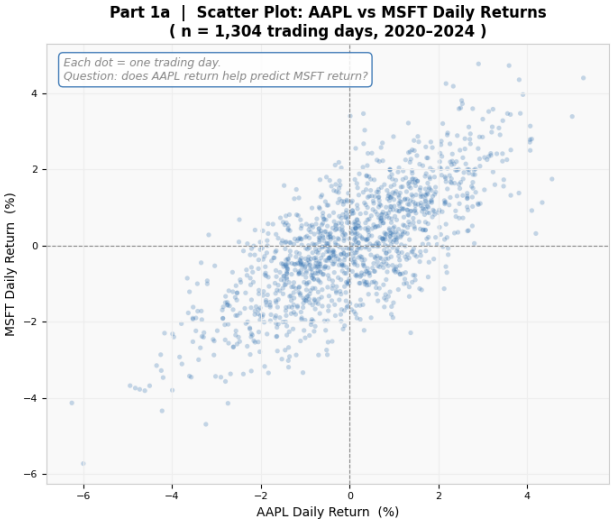
$$y = \alpha + \beta x + u$$

Example

$$\text{MSFT_ret} = \alpha + \beta\,\text{AAPL_ret} + u$$

- *x is an external variable; no explicit time index is required.*
- *Typical use: cross-sectional relation or static contemporaneous relation.*

Regression: $y \sim x$ Figure



Regression: $y_t \sim x_t$

Specification

$$y_t = \alpha + \beta x_t + u_t$$

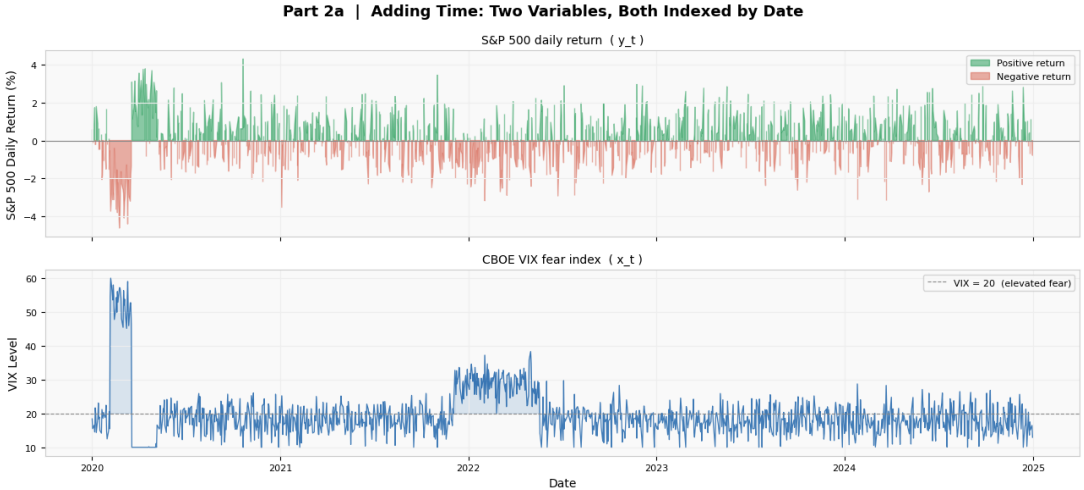
Example

$$\text{SP500_ret}_t = \alpha + \beta\,\text{VIX}_t + u_t$$

Variables in this expression

- VIX_t: Cboe Volatility Index (expected 30-day volatility of U.S. equity market implied by S&P 500 options).
- SP500_ret_t: one-period return of the S&P 500 Index.
- *x remains external, but both variables are time-indexed.*
- *Interpretation focuses on same-period comovement over time.*

Regression: $y_t \sim x_t$ Figure

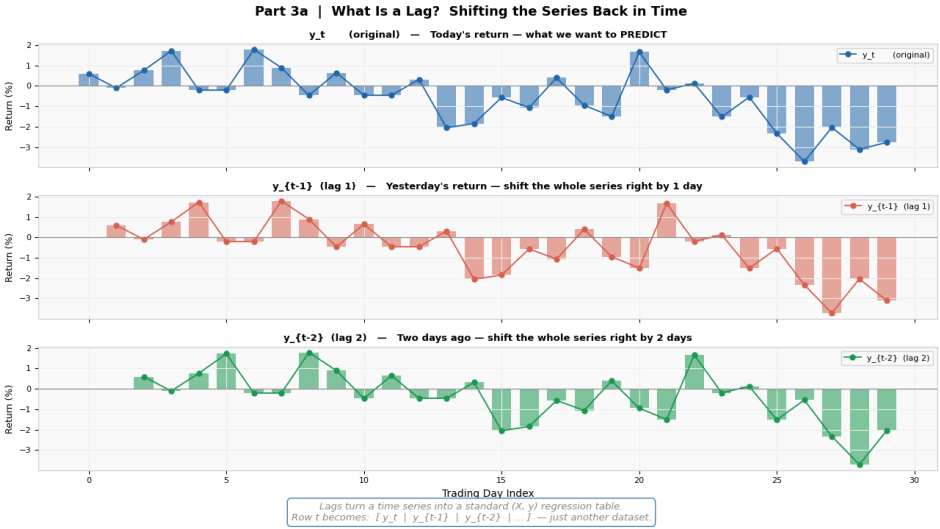


Both variables move through time together.
Question: can today's VIX (x_t) predict today's S&P 500 return (y_t)?

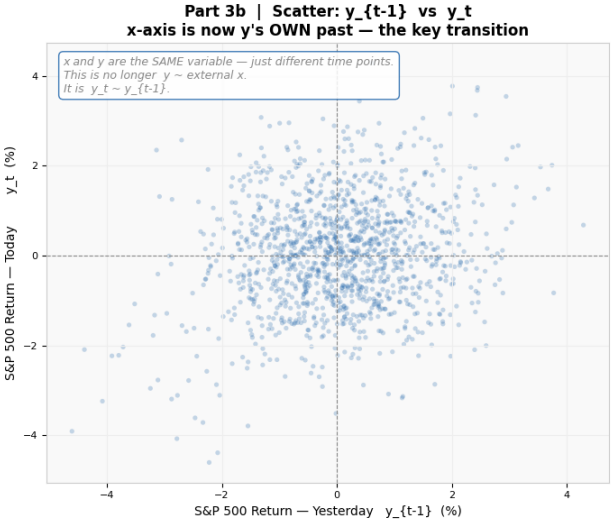
Lag Transition: From x_t to y_{t-1}

- A lag is a shifted copy of the same series.
- Replacing x_t by y_{t-1} turns regression into autoregression.

Lag Transition Figure 1



Lag Transition Figure 2



Autoregression: AR(1)

Specification

$$y_t = \alpha + \phi_1 y_{t-1} + u_t$$

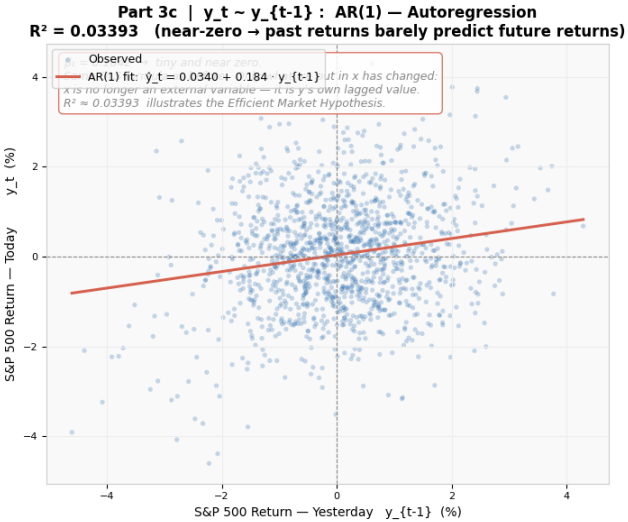
Example

$$\text{SP500_ret}_t = \alpha + \phi_1 \, \text{SP500_ret}_{t-1} + u_t$$

Variable note

- SP500_ret_t: one-period return of the S&P 500 Index.
- *The regressor is the lag of y itself.*
- This is the first autoregressive model: AR(1).

Autoregression AR(1) Figure



Autoregression: AR(2)

Specification

$$y_t = \alpha + \phi_1 y_{t-1} + \phi_2 y_{t-2} + u_t$$

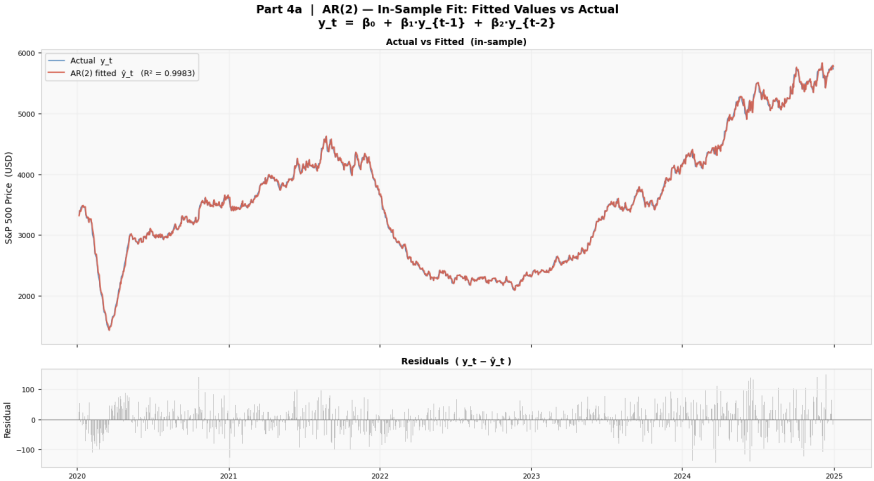
Example

$$\text{SP500_price}_t = \alpha + \phi_1 \, \text{SP500_price}_{t-1} + \phi_2 \, \text{SP500_price}_{t-2} + u_t$$

Variable note

- SP500_price_t: S&P 500 Index level at time t.
- *AR(2) extends persistence with two lags of y.*
- In this lecture context, treat it as an in-sample fit illustration.

Autoregression AR(2) Figure



High R^2 (0.9983) in-sample is expected: tomorrow's price ≈ today's price.
The residuals are what the model cannot explain — the gap between $\hat{y}_t$ and y_t.
What exactly IS that gap? That is the question we answer next:
→ Information sets, unobserved components, and conditional expectation.

Part A Checkpoint

1. What is the target variable and forecast horizon?
2. What information is available at forecast time?
3. Is the data structure suitable for forecasting tasks?

Baseline Forecasts: Why They Matter

- Baselines are simple benchmark models.
- If a complex model cannot beat baselines out-of-sample, it is not useful.

Naive forecast

$$\hat{y}_{t+1|t} = y_t$$

Mean forecast

$$\hat{y}_{t+1|t} = \bar{y}_t = \frac{1}{t}\sum_{i=1}^{t} y_i$$

Random Walk and AR(1)

Random walk

$$y_t = y_{t-1} + u_t$$

$$\Rightarrow \quad \hat{y}_{t+1|t} = y_t$$

AR(1)

$$y_t = c + \phi y_{t-1} + u_t$$

- If $\phi = 1$ and $c = 0$, AR(1) reduces to random walk.
- ϕ captures persistence in the series.

Moving Average Smoothing (Descriptive Tool)

$$MA_t = \frac{1}{m} \sum_{j=0}^{m-1} y_{t-j}$$

- m: moving window length.
- Use: smooth short-term noise and reveal underlying movement.
- This is a smoothing statistic, not yet an MA(q) estimation model.

Stationarity (Weak) and Why It Matters

A weakly stationary series has:

- Constant mean over time
- Constant variance over time
- Autocovariance depending only on lag, not calendar time

Many ARMA/ARIMA-style methods rely on approximate stationarity.

Autoregression AR(p) and Autocorrelation

AR(p) model

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + u_t$$

Autocorrelation at lag k

$$\rho_k = \mathrm{Corr}(y_t, y_{t-k})$$

- AR(p) uses internal history of y to explain and predict current value.

ARIMA Overview

ARIMA(p, d, q)

- AR: dependence on lagged values
- I: differencing to reduce non-stationarity
- MA: dependence structure in shocks

First difference operator

$$\Delta y_t = y_t - y_{t-1}$$

Model choice in practice:

- *Strong external predictors $\Rightarrow$ regression is often preferred.*
- *History-dominant series $\Rightarrow$ ARIMA is often preferred.*

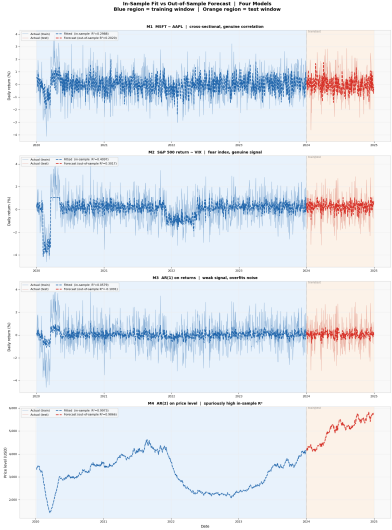
Model Assessment: In-Sample vs Out-of-Sample

- **In-sample fit**: how well a model explains training data.
- **Out-of-sample fit**: how well it predicts unseen future data.
- Forecasting decisions should prioritize out-of-sample performance.

Train/test logic

- Estimate model on training period only.
- Evaluate results on test period only.

Model Assessment Figure



Out-of-Sample Result from Horserace

- **M2 (SP500_ret VIX)** has the best out-of-sample R^2: 0.3017.
- **M1 (MSFT AAPL)** is second with out-of-sample $R^2 = 0.2020$.
- **M3 (AR(1) return)** has out-of-sample $R^2 = -0.1091$: worse than predicting the mean.
- **M4 (AR(2) price level)** has high out-of-sample R^2, but it is on price levels (different scale).

A model that looks good in-sample can still fail on future data.

Forecast Metrics: R^2

$$R^2 = 1 - \frac{\sum_{t=1}^{T}(y_t - \hat{y}_t)^2}{\sum_{t=1}^{T}(y_t - \bar{y})^2}$$

- Measures the fraction of variation in y_t explained by the model.
- Example: $R^2 = 0.70$ means about 70% of variation is explained.
- $R^2 < 0$ out-of-sample means the model is worse than predicting the mean.

How to Read R^2 in Forecasting

- In-sample R^2: fit to past data (can be inflated by overfitting).
- Out-of-sample R^2: real forecasting value on unseen data.

Model selection guidance:

- *High in-sample R^2 is not enough.*
- *A model is credible only when out-of-sample R^2 and errors are strong.*

Forecast Error Metrics (Out-of-Sample)

One-step-ahead forecast error

$$e_{t+1} = y_{t+1} - \hat{y}_{t+1|t}$$

MAE

$$MAE = \frac{1}{N}\sum_{i=1}^{N} |e_i|$$

Average miss per period, in the original unit.

RMSE

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N} e_i^2}$$

Similar to MAE, but large misses receive more penalty.

Forecast Error Metrics (continued)

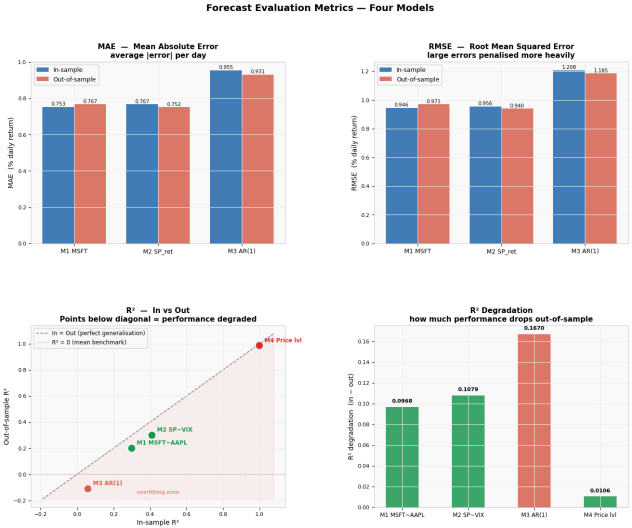
MAPE

$$MAPE = \frac{100\%}{N} \sum_{i=1}^{N} \left| \frac{e_i}{y_i} \right|$$

Average miss as a percentage of the true value.

- Easy to compare across scales.
- *MAPE is unstable when y_i is near zero (common in returns).*

Forecast Metrics Figure



Out-of-Sample Metrics: Reported Results

Model	MAE	RMSE	Out-of-sample R^2
M1: MSFT_ret $\sim$ AAPL_ret	0.767	0.971	0.202
M2: SP500_ret $\sim$ VIX	0.752	0.940	0.302
M3: AR(1) return	0.931	1.185	-0.109
M4: AR(2) price level	46.011	58.376	0.987

M2 generalizes best among return models, while M3 fails out-of-sample despite acceptable in-sample fit.

End-of-Lecture Checklist

1. Is the forecasting target and horizon clearly defined?
2. Are predictors valid under the information set at forecast time?
3. Does the proposed model beat simple baselines out-of-sample?

End